User Requirements

Emp/o is a third-party e-commerce pla8orm where merchants can sell products, shoppers can buy them, and adver/sers can promote selected products. The pla8orm also includes moderators and administrators responsible for maintaining service quality and enforcing pla8orm rules. Emp/o itself does not own or ship products. Payments are processed through Stripe, which creates transac/on records stored in the system.

Shoppers

Shoppers can browse available products using the search feature or by exploring the main and lis/ng pages. Each product includes a /tle, price, image, and descrip/on. Shoppers can either buy a product directly or add it to their cart for later checkout. The system prevents purchases that exceed available stock and no/fies users if a product is unavailable.

When a purchase is made, Stripe processes the payment and returns confirma/on to Emp/o. The corresponding transac/on details are stored in the Transac/on class, and the relevant merchant is informed to deliver the product to the shopper's shipping address. Shoppers can update their profile informa/on and report products that appear fraudulent or inappropriate. Reported products are reviewed by merchant moderators.

Merchants

Merchants manage their inventory through a dedicated panel that allows them to add, edit, or remove products. Each listed item includes details such as /tle, price, descrip/on, and image. Merchants can also update their business informa/on, including warehouse loca/on and bank account details. Once a shopper purchases a product, the merchant receives no/fica/on and fulfills the delivery independently.

Adver/sers

Adver/sers create campaigns that promote selected products on shopper pages. When crea/ng a campaign, the adver/ser chooses its name and adds one or more products. Each product added to a campaign requires payment of a fixed adver/sement fee. Payment is processed through Stripe, which generates a transac/on link and stores the transac/on informa/on in the system. Adver/sers can later add more products to an exis/ng campaign by paying addi/onal adver/sement fees through new transac/ons linked to the original one. Campaigns can be viewed and managed from the adver/ser's panel, where adver/sers may update campaign details or

remove included products. Moderators and Administrators Emp/o has two types of moderators: Support Moderators and Merchant Moderators. Support Moderators handle user issues and manage support /ckets, closing them when resolved. Merchant Moderators oversee reported products and can remove items or restrict merchant accounts if they violate pla8orm rules. Each modera/on type includes two permission levels: Regular Moderators handle standard reports and support requests. Admin Moderators serve as department heads who supervise their teams and can suspend members of their department. Adver/sing and Display System Products are displayed on shopper pages in a grid layout. Adver/sed products from ac/ve campaigns receive more visible placement, while regular items are shown according to availability and category. The display logic ensures a fair mix of sponsored and regular lis/ngs while maintaining the pla8orm's usabilityI'm

Emptio UC diagram

Emptio System

Update Profile
View Profile
Create Ticket
User
Search Product
Buy product
<<extends>>
<<extends>>
View Products
Add to cart
<<extends>>
<extends>
Report a Product
Purchase Cart
<<Includes>>
Transaction realisation
Shopper
View Cart
<<extends>>
Add a product
<<include>>
Edit Product
<<extends>>
View list of products
Merchant
Check the shoppers order
<<include>>
Stripe
Archive Inactive User
Every 2 years
Add a campaign
<<includes>>
Add Product to Compaign
Advertiser
View Compaigns
<<extends>>
View Reports
<includes>
Market Mod
Ban merchant
<<extends>>
Suprevise Solved payment
<<extends>>
Suspend market mod
Market Admin
View tickets
<<include>>
Issue archivation
Support
<<extends>>
Adjust issue archive
<<includes>>
Suprevise Solved issue Review
<<extends>>
Suspend support
Support Admin

Use Case Scenario -Add Product

Actor: Merchant

Purpose and Context: This use case allows the merchant to add new products to their inventory on Emptio. It enables merchants to expand their online catalog by entering product details such as name, description, price, category, quantity, and images.

Assuptions:

1. The Merchant has all the required product information (name, price, description, stock quantity.

Pre-Conditions:

1. The Merchant's account is verified and active

Initiating Business Event(Trigger): 1. The Merchant clicks on the **"Add Product"** option in the merchant panel.

Basic Flow of Events:

1.The Merchact press on the "Add Product" option in the merchant panel
2.The System displays the Edit Product page with a form to input product details
3.The Merchant enters all required product information - name, price, category, description, quantity, and upload images
4.The System validates the entered data
5.The Merchant clicks "Submit" button
6. The System saves the product details in the database and adds the product to the merchant's list
7.The System displays success message - "Product added successfully"
8.The Merchant views the new product in inventory list

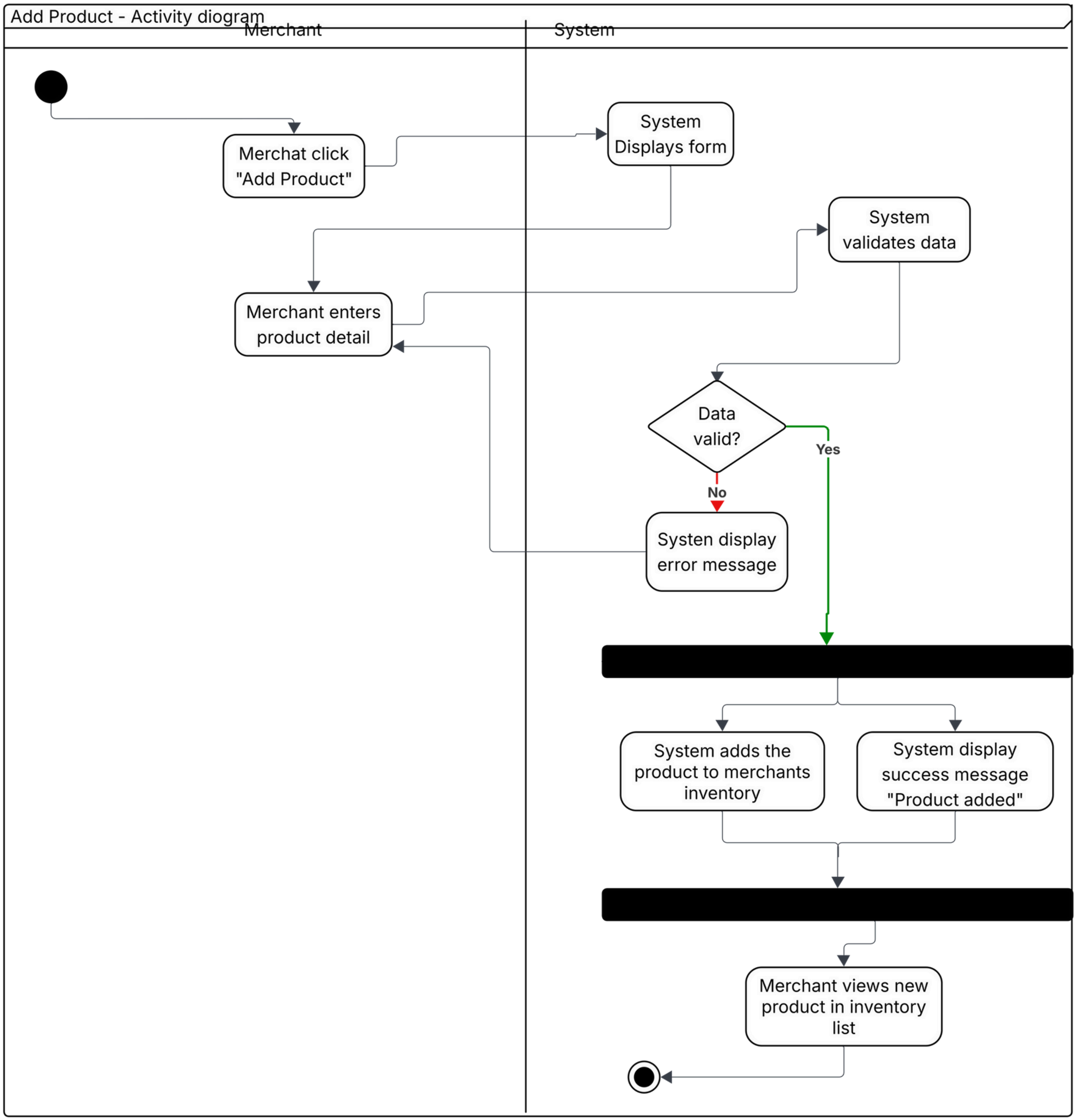
Alternative Flow of Events:

Missing Required Fields:

4A1. The System detects that one or more required fields are empty or contain invalid values
4A2. The System display an error message specifying which fields must be corrected.
4A3. The Merchant corrects the errors and resubmits the form

Post-Condition:

1.The new product is successfully added to the merchant's inventory and becomes visible in Inventory page
2.The system updates the product database with new entry


Add Product - Activity diogram
Merchant
System
Merchat click "Add Product"
System Displays form
System validates data
Merchant enters product detail
Data valid?
Yes
No
Systen display error message
System adds the product to merchants inventory
System display success message "Product added"
Merchant views new product in inventory list

Use Case Scenario - Add Campaign

Actor: Advertiser

Purpose and Context: The use case is used by the advertiser to create a new advertising campaign. It allows the advertiser to define campaign details such as name, description, duration, and linked products, so that the campaign can be activated and managed within the system.

Assumptions:
1.The advertiser has the necessary permissions to create new campaigns.
2.The products intended to be included in the campaign already exist in the system.
3.The campaign name chosen by the advertiser is unique.

Pre-Conditions:
1.The advertiser has access to the campaign management panel.
2.The advertiser has at least one active product available to associate with the campaign.
3. The system has campaign creation functionality active and accessible

Initiating Business Event(Trigger): 1. The advertiser clicks on the **"Add Campaign"** button in the campaign management panel.

Basic Flow of Events:
1.The advertiser clicks on the **"Add Campaign"** button to start creating a new campaign.

2.The system displays the campaign creation form.

3.The advertiser enters the campaign name, description, start date, and end date.

4.The system records the entered information temporarily.

5.The advertiser selects one or more products to associate with the campaign.

6.The system displays a list of available products.

7.The advertiser confirms the selected products.

8.The system validates all entered campaign details and selected products.

9.The advertiser clicks **"Create"** to confirm the campaign creation.

10.The system creates a new campaign record linked to the advertiser and selected products.

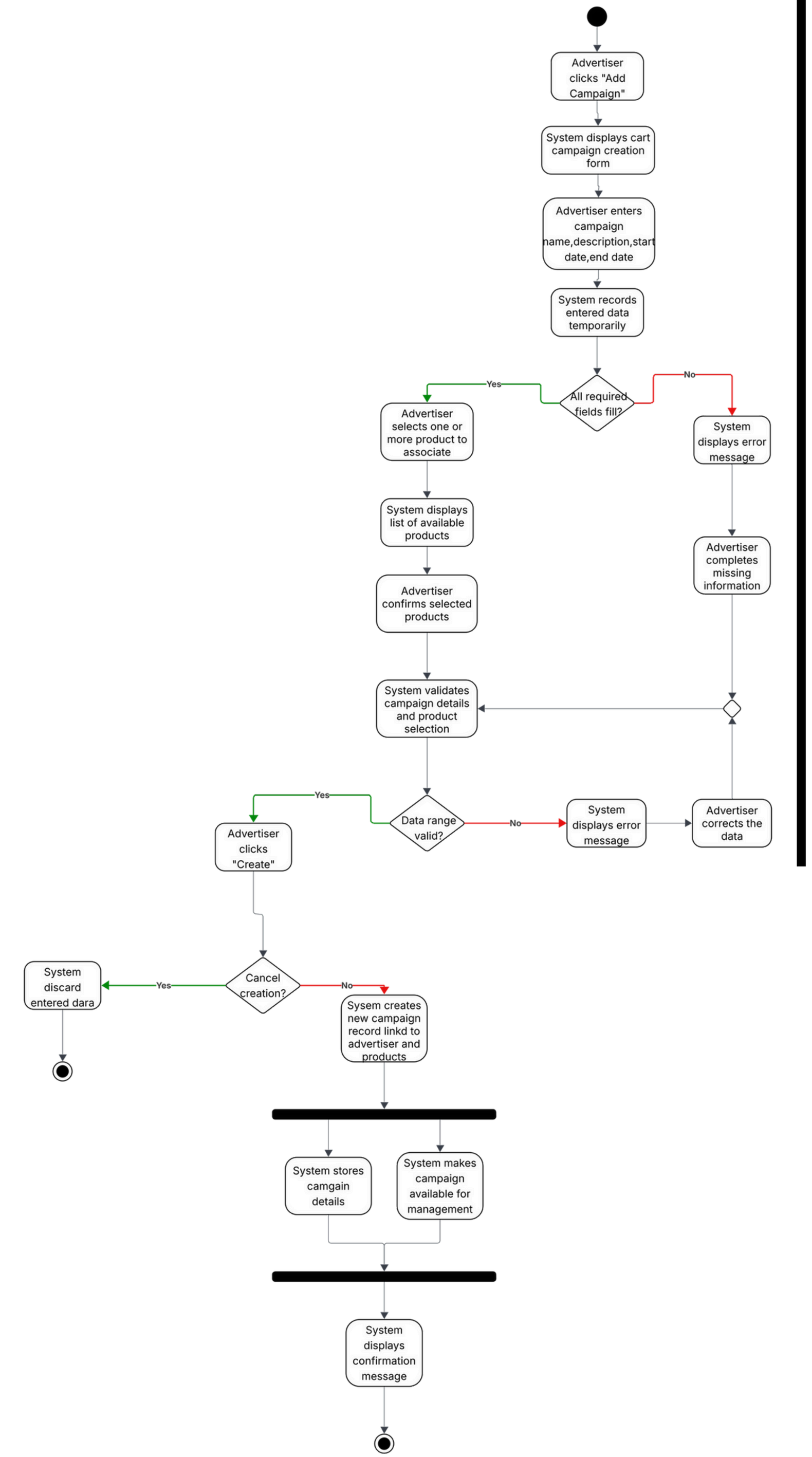
11.The system displays a confirmation message indicating the campaign was successfully created.

Alternative Flow of Events:
Missing required information
3A1. The advertiser leaves one or more required fields (e.g., campaign name or dates) empty.
3A2. The system displays an error message prompting the advertiser to fill in all mandatory fields.
3A3. The advertiser completes the missing information.
3A4. The flow returns to step **4** of the basic flow

Invalid data range
8A1. During validation, the system detects that the campaign end date is earlier than the start date.
8A2. The system notifies the advertiser that the date range is invalid.
8A3. The advertiser corrects the dates.
8A4. The flow returns to step **8** of the basic flow.

Campaign creation cancelled
Alternative scenario: Campaign creation canceled
9A1. The advertiser decides to cancel before confirming campaign creation.
9A2. The system discards all entered data.
9A3. The use case ends.

Post-condition:
1.The system creates a new campaign record linked to the advertiser and the selected products.
2.The system stores all campaign details, including name, description, and duration.
3.The campaign becomes available for further management or activation by the advertiser.
4.If the advertiser cancels the creation, no campaign record is created.


Advertiser clicks "Add Campaign"
System displays cart campaign creation form
Advertiser enters campaign name,description,start date,end date
System records entered data temporarily
All required fields fill?
Yes
No
Advertiser selects one or more product to associate
System displays error message
System displays list of available products
Advertiser completes missing information
Advertiser confirms selected products
System validates campaign details and product selection
Data range valid?
Yes
No
System displays error message
Advertiser corrects the data
Advertiser clicks "Create"
Cancel creation?
Yes
No
System discard entered dara
Sysem creates new campaign record linkd to advertiser and products
System stores camgain details
System makes campaign available for management
System displays confirmation message

Use Case Scenario "Purchase Cart"

Actor: Shopper

Purpose and context: A shopper adds needed items to the cart and after completing that proceeds to payment and delivery options.

Assumption: 1. The user is satisfied with the selected cart items also with their number, prices
.
Precondition: 1. The user is viewing the cart page with the list of selected items for PURCHASE.
2. There should be at least one item in the cart
.
Basic flow of events:
1. The shopper clicks on "Proceed to payment" button

2. "Select delivery address" page
   a. the shopper already has an information of delivery address in the user account and selects it
   b. In the other case the shopper fills the form correctly in the page of delivery address

3. Then payment page is shown , where the shopper selects a preferred payment method (not by cash)

4. After selecting that payment method generates a link for a transaction

5. After a successful purchase the message about it is displayed,**the check is sent to the email** and a link to track the order status,also it clears the cart of the shopper

Alternative flow of events:

Product in the cart are unavaliable
2b1. The system displays the list of such products and return to the main page

The payment is failed
4a1. The system displays the info and a button "Retry", the transaction link is regenerated

Post condition:
Log info: Update on history of transactions
Stock: Some items will be reduced or even out of stock

Purchase Cart

Shopper

System

View cart

Proceed to payment

All product avaliable

No

Message "Not all products in cart are avaliable"

Display a list of unavaliable products

Yes

Show available payment methods

Yes

Payment is failed. Try again?

No

No

Generate a link for transaction

Paid

Yes

Message"Success"

Present a link for order tracking

Clear Cart

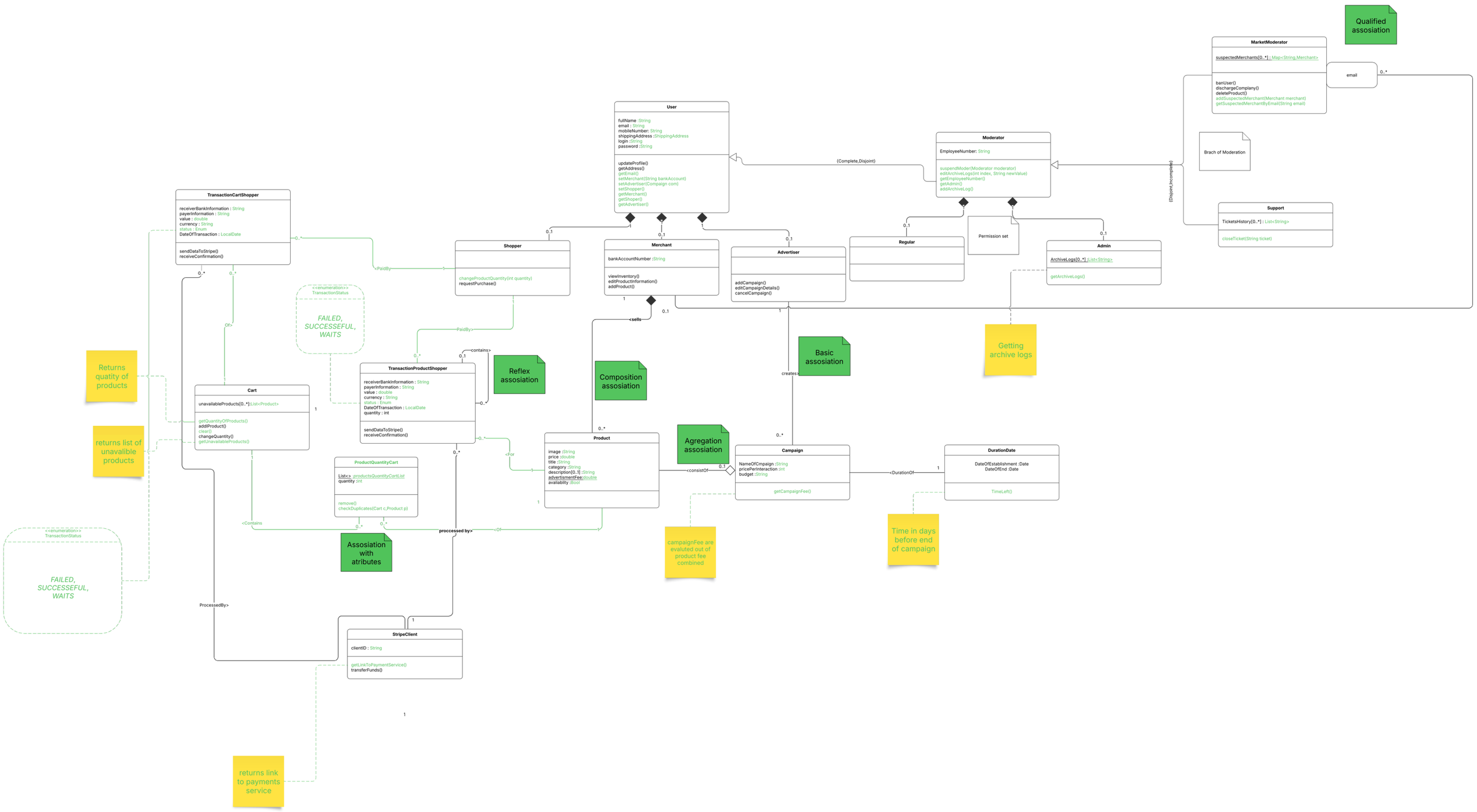
Open the link?

Yes

Show the order status

No

Back to the main page


Qualified assosiation
MarketModerator
suspectedMerchants[0..*] : Map<String,Merchant>
banUser()
dischargeComplany()
deleteProduct()
addSuspectedMerchant(Merchant merchant)
getSuspectedMerchantByEmail(String email)
email
0..*
Brach of Moderation
(Disjoint,Incomplete)
Support
TicketsHistory[0..*] : List<String>
closeTicket(String ticket)
User
fullName :String
email : String
mobileNumber: String
shippingAddress :ShippingAddress
login :String
password :String
updateProfile()
getAddress()
getEmail()
setMerchant(String bankAccount)
setAdvertiser(Compaign com)
setShopper()
getMerchant()
getShoper()
getAdvertiser()
(Complete,Disjoint)
Moderator
EmployeeNumber: String
suspendModer(Moderator moderator)
editArchiveLogs(int index, String newValue)
getEmployeeNumber()
getAdmin()
addArchiveLog()
Permission set
Regular
Admin
ArchiveLogs[0..*] :List<String>
getArchiveLogs()
Getting archive logs
TransactionCartShopper
receiverBankInformation : String
payerInformation : String
value : double
currency : String
status : Enum
DateOfTransaction : LocalDate
sendDataToStripe()
receiveConfirmation()
<PaidBy
Shopper
changeProductQuantity(int quantity)
requestPurchase()
Merchant
bankAccountNumber :String
viewInventory()
editProductInformation()
addProduct()
Advertiser
addCampaign()
editCampaignDetails()
cancelCampaign()
<sells
creates>
Basic assosiation
Composition assosiation
<<enumeration>>
TransactionStatus
FAILED, SUCCESSEFUL, WAITS
PaidBy>
contains>
Reflex assosiation
TransactionProductShopper
receiverBankInformation : String
payerInformation : String
value : double
currency : String
status : Enum
DateOfTransaction : LocalDate
quantity : int
sendDataToStripe()
receiveConfirmation()
Returns quatity of products
Cart
unavailableProducts[0..*]:List<Product>
getQuantityOfProducts()
addIfProduct()
clear()
changeQuantity()
getUnavailableProducts()
returns list of unavalible products
ProductQuantityCart
List<> productsQuantityCartList
quantity :int
remove()
checkDuplicates(Cart c,Product p)
<For
Product
image :String
price :double
title :String
category :String
description[0..1] :String
advertismentFee:double
availabilty :Bool
Agregation assosiation
<consistOf
Campaign
NameOfCmpaign :String
pricePerInteraction :int
budget :String
getCampaignFee()
campaignFee are evaluted out of product fee combined
<DurationOf
DurationDate
DateOfEstablishment :Date
DateOfEnd :Date
TimeLeft()
Time in days before end of campaign
<Contains
proccessed by>
<Of
Assosiation with atributes
<<enumeration>>
TransactionStatus
FAILED, SUCCESSEFUL, WAITS
ProcessedBy>
StripeClient
clientID : String
getLinkToPaymentService()
transferFunds()
returns link to payments service

Dynamic Analysis on “Add Product”,"Add compaign","Purchase Cart" use cases & “Cart class” state diagram.

Following the above-presented diagrams, a dynamic analysis was performed. The result of this analysis indicated a need for further expansion on the design class diagram to encompass newly identified requirements and behaviors of the system. All introduced changes can be seen in the below presented final design class diagram.

“Cart class” state diagram.

1.First of all transaction might be failed so we need to add a status of transaction in transaction class
2.Card is cleared every time shopper use perchaseCart() method seccussfuly so we need to add a method to clear the card so the shopper can cleared it if he wanted without buying the whole thing it might be a good method to make a money but not quite handy and fair.
3.Since in our card might be unavaliable products we need to show and evaluate which products are unvaliable in the card in do it dinamicly so we gonna add filed like /unavalibleProducts do display it to the shopper in case he want buy a cart and his cart contain such unavalible products

“Add Product”

No comments its gonna work with implementation we have now at least i hope its gonna work.

"Add compaign"

1.Compaign now have validation date and date of decloration and date of end which is logical but with simple implementation that is unachivable we need to add class Duration class for assosiating it with the compaign since its gonna have field like dateOfDecloration,dateOfEnd or smth similar and evaluating the gap between to know then compaign needed to be deleted

"Purchase Cart"

1 Have pretty much same problems as Cart class in general and they already described

/UnavalibaleProduct-its list of products which is curently in cart and have unvaible status.
Method clear() - clears out the cart out of products
Status in transaction is reffers to is transaction failed seccesful or waiting to be paid
DurationDate is new class explicetly for campaign its contain startDate and EndDate of the compaign also it evalute dimanicly how much time this compaign left in the future might be used for other implementations

The presented class diagram illustrates the changes necessary to model UML constructs that do not exist in the Java language, The changes introduced are highlighted in green. The following section will describe the transformation process of certain UML constructs such as attributes types.

Runtime validation logic:

Using the java custom runtime exception and basic if checks we making validation we could use Java Bean valotion for same purposes and make it faster and simple and more like a c# .net but we dont what u goona do about it

Optional Attributes:

Atributes that may coontain no value they are represented in UML as atribute with [0..1] notaition and inplemented as null values in java with separate constructor so if dont wanna create one u just dont type on while constacting the object examples: Product:description[0..1] :String

Multi-value atributes:

Java is nativly supporting Multi-value atributes as classes inherited from Colection class we using for our purposes ArrayList<E> from java.util this its worth mentioning how and what for we using that collection first of all to store in our customers cart a peroduct that he added there and potentionaly willing to buy we used arraylist generic nature to fill our needs in diffrenct scenarious we save custom class products in our card using that implementation as well as we ussing more in basic wawy in our moderator system which can write and adjast tickets examples:**Cart**-unavailableProducts[0..*]:List<Product>,**Admin**-ArchiveLogs[0..*] :List<String>,**MarketModerator**-suspectedMerchants[0..*] : Map<String,Merchant>,**Support**-TicketsHistory[0..*] : List<String>

Derived Atributes

Derived atributes its such atributes which is not located as field in our implemetation and rather count on the run in java we used regualr get methods as in ecapsulation principle to simulate existing of these atributes for example;TimeLeft() so

```
public int TimeLeft() {
    return dateOfExpiration.getDays() -
dateOfEstablishment.getDays();
}
```

Enums(tagged values)

We use it Transaction state in transaction class to deffrinciate from one another and base behaviour of our system based on it we use enums or enuminators for that purposes which is represented in java based libary

```
enum Status implements Serializable{
    FAILED,
    SUCCESSEFUL,
    WAITS
}

public class Transaction implements Serializable{
```

Complex atributes(class atributes)

Complex atributes its such atributes that are contain more then one simple field or could have more complex logic in they behaviour compare to the basic atrubites and field we implemented it using standart java tools they represented as inner classes of the class which have such complex atribute as an example i have here Date class which is inner class of DurationDate and it usses two fields Date one called start of seccond called end and they represent day motyj hour and year

```
public static class Date implements Serializable {
    private int year;
    private int month;
    private int day;
    private int hour;


    Date(int year, int month, int day, int hour) {
        if (month < 1 || month > 12 || day < 1 || day > 31 || hour < 0
|| hour > 23) {
            throw new InvalidFormatException("Invalid date format");
        }
        this.year = year;
        this.month = month;
        this.day = day;
        this.hour = hour;
    }


public class DurationDate implements Serializable {
    private static final List<DurationDate> extent = new
ArrayList<>();
    private static final String EXT_FILE = "duration_extent.ser";


    private Date dateOfEstablishment;
    private Date dateOfExpiration;
```

.

Extension classes(how we saving our progress):

Since using databasesis forbiden and some misterious wispers from within the reality dectates us to use an extension classes to save our systems basicly it method of saving objects in the file and initilazie it later in even completly another session of the programs and what was i describing was a Serialization and particulary waht we using in java for such puprposes is Serializable interface which of our classes inherit from to get that functionality also we have a static field which is represented with object of ArayList class which as we remember is a collection and everytime constractor of the class are triggered ArayList get filled up with our class and every point of them system all objects of the specific class can be saved in thorugh a static method also we using inharitance becouse by making that functionality in User all of its child classes like Merchant ,Advertiser,Shopper already have it implementation but its formality we can from any point we wont use any additional frameworks all of this just basic java so that it is how it is

Tests:

Here avtually we using the additional functional to make test we using JUnit testing for java as tool to create test easier and faster and more be more precised in our testing so we test our i would called straight forward normal people called dumb vilation of the atributes for every class we implemented and as well test testing fow our class seralization work or how our Extension calsses system work

General assosiation implementation

AddMethod(): In most of our relations the direction is reversed, so the add method of one object calls the add method of another object. To avoid looping, every connected add method has an if-statement that checks whether the relation already exists in the container. If it does, the process stops.

RemoveMethod(): Basically the same but in reverse. The removal process can be triggered from either side, and it automatically removes the bound in the opposite object. This is controlled with an if-statement check.

EditMethod(): We don't have many examples of this, but it works in reverse in our case. For example, a Product cannot modify itself, but a Merchant can modify it through setters in the Product class. These setters check whether the Product belongs to the Merchant performing the modification.

Composition Association: Represented in our project as Merchant and Products. Products cannot exist without the Merchant who sells them. Since we are implementing everything in Java, we add a method to delete a Merchant. It goes through the list of associations and deletes all bounds in both the Merchant and the Products created by it. Because Java does not allow manual deletion of objects, the idea is that without references the garbage collector should delete them, but I am not fully sure about that.

Qualified Association: Implemented as a bound between a Supervisor (MarketModerator object) and a Merchant in the list of suspected merchants. It takes the form of a map where the key is the email and the value is the Merchant object. Since the Merchant map is static, any moderator can access it, but each Merchant also keeps a record of which moderator added them to the suspected container. Since a Merchant can change its email at any time, the map updates to the Merchant's new email. That is basically how it works.

Association Class: Represented as ProductQuantityCard. Every time a Product is added to the Cart, it uses the constructor of that class, which requires a Cart object, a Product object, and an integer quantity. It can be created from both the Product and the Cart, which is not the most logical thing, but it works in reverse and functions properly. It also has a unique feature I implemented during a sleepless night: instead of throwing an identity exception, it checks if the same Product was added to the same Card before, regardless of quantity. If it was, it adjusts the existing relation and rewrites the quantity to be the sum of the old and new values. I thought it was cool, even if nobody will read the description or code.

Reflex Association: As an example of reflex association, we use the Transaction class. It has a child field and a parent field, and the addChild method is reversed because it all comes from the same class, and both objects can use it. It is more logical to add children to the original object rather than the other way around, but still, when you add a child transaction, the child adds you back as the parent. The difference is that the child is a container, and the parent is a single field represented in the object. It is three in the morning and I am writing this on the go, so I might have misinterpreted something.

Association Validation: Done with a basic if-check to see if the container already holds the relation, which breaks the loop. As mentioned above, the association class also performs an individuality check, but instead of stopping the method or firing an exception, it adds the new quantity to the original one. Since there is no way to stop a constructor, it creates an empty object not tied to any relation, which is instantly removed by the Java garbage collector. Some associations do not need a check (e.g., reflex, qualified), but we still check for null objects. Reverse connections are often done through recursion, where the add method of one class calls the add method of the other and vice versa. I already explained how we prevent endless looping. To sum up our implementation: we used the provided documents and tried to keep everything as simple and functional as possible.

Inheritances(how we connect classes without proper tools):

Mostly by using Composition we only have too cases there we had to impelement this compulsory inheritance architecture first on our operational table is

Multi-aspect inheritance: Briefly, to bring you up to speed: we have moderators in our system, which are divided into *market moderators* and *support moderators*. Both types can be heads of their respective branches and have access to specific logs and features, such as the ability to suspend moderators. Essentially, they act as supervisors. We implemented this using the composition approach, which was the most suitable solution for us considering the number of relationships and classes involved it's like a web. Serializing that web alone would be a headache. As for the implementation yes, it might seem like a questionable choice if you think about it twice, but I didn't even think once. We placed all unique moderator methods in the Moderator class, and they can be used only if the associated Admin object is not null. Obviously, we don't want anyone to create a moderator without an admin. This was done for the sake of generalization: instead of duplicating the same functionality across inherited classes, we centralized it in the Moderator superclass. The logic is identical anyway. So, we have the Moderator class acting as the superclass (or we can say "father class") of both MarketModerator and SupportModerator. The Moderator class contains a field that holds a reference to an Admin. This reference grants access to admin-specific methods originally tied to the Admin class. If the reference is null, moderator-admin functionality is simply unavailable, since the log list exists only inside the admin object and is static. Basically, that's the idea. We establish the connection using the constructor.

Overlapping inheritance: This case is not unique composition was used again for the same reasons that flattening was not. It would be even worse in this situation: imagine rewriting all the small tests we have. There are 126 of them, and many of them use multiple constructors within a single test. So, what we have here is a User as the parent, and Moderator, Shopper, and Advertiser roles, which now cannot exist without the User, because they can only be created inside the User object via setter methods. For example, setMerchant collects additional data (like the bank account), which is crucial for constructing a Merchant. The Merchant then stores it as its own field and forms a reversed connection to the User.
This was needed because we had some relations that went beyond what simple inheritance with extends could handle, especially since some relationships required access to fields in Merchant that originally belonged to User. In short: a User contains (or does not contain) exactly one Merchant, Advertiser, or Shopper. They cannot exist without a User, and therefore they store a reference back to the User. If we need anything from them, we simply approach the parent User and get the required information essentially, the User is an overprotective father to its daughters, if you want an analogy.
Most of the exception handling stayed the same, because we are still using trimmed versions of the original class constructors. So there is nothing to worry about regarding the tests in general they were implemented mostly for multi-aspect behavior. The number of tests I already rewrote covers all features that could be applied. I suppose I can add a test where I try to add multiple roles to a single User, but this case is already included in the existing tests I just need to look more carefully.